

SURAT THANI, Thailand

WMO: 485510

Lat: 9.1335N

Lon: 99.1521E

Elev: 23

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	69.1	70.5	65.3	93.6	79.6	67.0	99.2	79.0	19.0	82.6	17.6	82.6	0.4	330	0.303

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.7	96.5	78.9	94.9	79.1	93.5	79.1	82.5	89.7	81.7	89.3	81.0	88.9	7.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	160.4	85.6	80.0	156.0	85.3	79.1	150.9	84.9	46.5	88.7	45.5	90.3	44.8	89.3	92.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	14.5	12.7	DB	65.5	99.3	2.4	2.3	63.8	101.0	62.4	102.4	61.1	103.7	59.4	105.4
			WB	63.6	85.5	2.9	2.9	61.5	87.5	59.8	89.2	58.2	90.9	56.0	93.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	79.3	80.5	82.8	84.6	84.3	83.4	82.8	82.6	82.3	81.5	80.3	79.1
	DBStd	2.59	1.87	1.81	1.92	2.02	2.12	1.95	1.76	1.87	1.77	1.82	1.92	2.06
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11668	910	855	1018	1039	1064	1001	1015	1010	968	977	908	903
	CDD65	6193	445	435	553	589	599	551	550	545	518	512	458	438
	CDH74	59239	3812	4327	6023	6687	6192	5543	5258	5103	4599	4354	3825	3516
CDH80	23606	1286	1847	2885	3275	2678	2280	2092	2027	1722	1450	1107	957	
Wind	WSAvg	3.9	4.5	4.6	4.4	3.6	3.4	4.0	4.2	4.2	3.5	3.0	3.5	4.5
Precipitation	PrecAvg	72.1	2.1	0.9	3.1	3.2	7.1	6.3	7.0	6.8	8.5	9.9	11.0	6.2
	PrecMax	91.9	11.1	6.3	30.3	9.0	10.1	10.0	11.0	18.0	14.5	25.1	23.5	21.6
	PrecMin	53.9	0.0	0.0	0.0	0.1	3.2	3.1	3.1	2.9	3.7	4.3	3.2	1.9
	PrecStd	9.7	2.3	1.4	5.7	2.4	2.0	2.0	2.2	2.9	2.9	4.4	5.8	4.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.3	93.5	97.0	99.2	98.7	95.3	95.0	95.1	93.9	93.1	91.4	89.9
		MCWB	77.5	77.0	77.2	79.1	79.4	79.5	79.0	79.0	79.7	79.3	79.4	78.8
	2%	DB	89.4	91.6	94.9	96.9	95.4	93.6	93.0	93.2	92.4	90.9	89.4	88.1
		MCWB	77.3	77.1	77.8	78.9	80.2	79.4	79.0	79.0	79.5	79.4	79.2	78.5
	5%	DB	87.9	90.0	93.3	95.0	93.4	92.2	91.4	91.4	90.8	89.3	87.8	86.8
		MCWB	77.1	77.1	78.0	79.3	80.2	79.3	78.8	78.7	79.0	79.2	79.1	78.0
	10%	DB	86.4	89.1	91.6	93.1	91.5	90.6	89.6	89.5	88.7	87.5	86.1	85.3
		MCWB	76.6	77.0	77.8	79.2	80.0	79.3	78.7	78.6	78.9	79.1	78.5	77.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.4	80.6	81.8	82.7	83.5	83.1	82.0	81.9	82.2	82.4	82.4	81.8
		MCDB	86.9	88.4	90.2	91.7	91.6	90.0	88.4	89.3	88.2	87.7	86.2	85.9
	2%	WB	79.2	79.2	80.4	81.5	82.2	81.7	80.9	80.8	80.8	80.8	80.7	80.1
		MCDB	85.9	87.6	89.7	91.0	90.5	89.2	88.4	88.6	88.5	87.4	86.5	85.5
	5%	WB	78.5	78.6	79.6	80.8	81.3	80.8	80.1	80.0	80.0	80.0	79.8	79.1
		MCDB	85.1	87.1	89.0	90.7	89.7	88.6	87.8	87.8	87.7	86.6	85.4	84.4
	10%	WB	77.6	77.8	78.9	80.0	80.5	79.9	79.4	79.3	79.2	79.2	79.0	78.2
		MCDB	84.1	86.3	88.4	89.7	88.9	87.5	86.9	86.8	86.5	85.5	84.4	83.2
Mean Daily Temperature Range	5% DB	MDBR	14.8	17.9	19.1	18.7	16.5	15.5	15.2	15.3	14.9	13.5	11.7	12.0
		MCDBR	17.2	19.6	21.4	21.2	19.0	18.0	17.6	18.1	17.4	16.0	14.9	15.0
	5% WB	MCWBR	7.3	7.8	7.4	6.6	6.7	6.5	6.3	6.6	6.7	6.4	6.7	6.9
		MCWBR	15.2	18.3	19.7	19.5	17.6	16.3	16.1	16.2	16.1	14.7	13.5	13.2
Clear-Sky Solar Irradiance	taub		0.481	0.491	0.498	0.475	0.447	0.444	0.445	0.437	0.427	0.446	0.471	0.481
	taud		2.197	2.161	2.177	2.247	2.368	2.411	2.392	2.404	2.424	2.357	2.273	2.214
	Ebn at Noon		264	265	264	267	269	267	268	274	280	275	265	261
	Edh at Noon		46	49	49	45	39	37	38	38	38	40	43	45
All-Sky Solar Radiation	RadAvg		1485	1779	1847	1823	1573	1501	1466	1501	1494	1380	1285	1273
	RadStd		132	114	160	107	105	87	98	97	87	127	116	125

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (15 neighbors)	+0.45	+0.70	+1.08	+1.42	+1.03	+1.19	N/A	N/A	+220	+220
	+0.34	+0.63	N/A	N/A	+0.36	+0.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page